

Yanlong Yang

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RESEARCH INTERESTS

Generative models (diffusion) for inverse problems and computational imaging; world models; multi-sensor fusion (radar, LiDAR, camera) for autonomous-driving perception; 3D object detection, tracking, and point-cloud enhancement; radar signal processing.

EDUCATION

- James Cook University** Apr 2024 – Present
Ph.D. in Engineering — Cairns, Australia
— Advisor: Dr. Tao Huang. Generative priors for radar super-resolution point-cloud generation; extending toward world models for perception.
- Beijing Institute of Technology** Sep 2017 – Jul 2020
M.Eng. by Research, Optical Engineering — Beijing, China
— Thesis: 3D Human Thermal-Surface Reconstruction Technology and System (86.8/100). Advisor: Dr. Chao Xu.
- Tianjin University** Sep 2013 – Jul 2017
B.Eng., Opto-Electronics Engineering — Tianjin, China
— Honours thesis: phase transition of $\text{Ba}_{0.6}\text{Sr}_{0.4}\text{TiO}_3$ -silicon hybrid metamaterial by THz spectra (86/100). Advisor: Dr. Liang Wu.

PUBLICATIONS

* denotes equal contribution / co-first author.

Peer-reviewed journals

- RaLiBEV: Radar and LiDAR BEV Fusion Learning for Anchor-Box-Free Object Detection Systems.** Y. Yang*, J. Liu*, T. Huang, Q.-L. Han, G. Ma, B. Zhu. *IEEE Transactions on Circuits and Systems for Video Technology*, 35(5):4130–4143, 2025.
- Fusion Reconstruction Method for 3D Temperature Fields on the Human Body Surface.** Y. Yang, C. Xu. *Infrared Technology*, 2022.
- Observation of Phase Transitions of $\text{Ba}_{0.6}\text{Sr}_{0.4}\text{TiO}_3$ -Silicon Hybrid Metamaterial by THz Spectra.** L. Wu, Y. Fu, N. Xu, Q. Sheng, X. Zhang, Y. Yang, et al. *ACS Applied Electronic Materials*, 2020.

Peer-reviewed conferences

- Self-Diffusion Driven Blind Imaging (DeblurSDI).** Y. Yang, G. Luo. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- SSP: Small Object Detection from Sparse Point Clouds of Roadside LiDAR for Vehicle-to-Infrastructure Cooperating Systems.** H. Liang, Y. Yang, J. Li, Y. Deng. *ITS World Congress*, 2023.

Preprints & under review

- Self-Diffusion for Solving Inverse Problems.** G. Luo, S. Huang, Y. Yang. arXiv:2510.21417, 2025.
- Unsupervised Radar Point-Cloud Enhancement via Arbitrary LiDAR-Guided Diffusion Prior.** Y. Yang, J. Liu, G. Luo, H. Li, E. Ahn, M. Rahimi Azghadi, T. Huang. arXiv:2505.09887, 2025.

RESEARCH & PROFESSIONAL EXPERIENCE

- James Cook University — Doctoral Researcher** Apr 2024 – Present
— Designing and implementing an end-to-end method that generates LiDAR-density point clouds from raw vehicle-mounted radar signals using generative/diffusion priors.
— Extending the work toward world models for autonomous-driving perception.
— Contributed to an AI crocodile-detection system (JCU – Queensland Government / DETSI collaboration): computer-vision models trained on thousands of hours of footage to distinguish crocodiles from other floating objects, deployed on a mobile trailer-mounted camera platform for real-time public-safety monitoring in North Queensland (2026).
- Vanjee Technology, Beijing — Algorithm Engineer** Jul 2020 – Mar 2024
— Radar–LiDAR fusion for robust object detection in adverse weather (basis of RaLiBEV).
— Led a 3-person team building a data closed-loop system for vehicle–road cooperative autonomous driving; shipped v1.0 to customers.

- Built a MIMO radar signal-processing toolkit, deployed on engineering-sample hardware.

Beijing Institute of Technology — Graduate Researcher

Sep 2018 – Jul 2020

- Designed the hardware, software, and mechanical system for human 3D thermal-surface reconstruction; results published in *Infrared Technology*.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB; PyTorch, PCL, OpenGL, Qt.

Tools: Git, LaTeX, COMSOL, Altium Designer, Visio, MS Office.

Domains: Diffusion/generative modeling, inverse problems, sensor fusion, 3D reconstruction, digital signal & image processing, autonomous-driving perception.

AWARDS & LANGUAGES

- 2nd place (24th of 200 teams), NXP Cup — China National College Students' Smart Car Competition, 2017.
- Languages: Chinese (native); English (IELTS 6.5, 2023).

REFERENCES

Dr. Tao Huang — Director, Intelligent Computing & Communications (ICC) Lab, James Cook University, Australia · tao.huang1@jcu.edu.au · *Ph.D. advisor*

Prof. Weiqi Jin — MOE Key Lab of Optoelectronic Imaging Technology & System, Beijing Institute of Technology, China · jinwq@bit.edu.cn · *Master's advisor*

Prof. Liang Wu — Key Lab of Opto-Electronics Information Technology, Tianjin University, China · wuliang@tju.edu.cn · *Bachelor's advisor*